

What regulatory, payment policy, or programmatic design changes should HHS prioritize?

1. Reduce Information Asymmetry to Increase Competition

Information asymmetry stifles competition in health care. Large entities increasingly acquire, interpret, control, and restrict access to diagnostic data — especially in radiology imaging. Diagnostic test acquisition, interpretation, and data control are vertically integrated.

HHS should prioritize policies that reduce information asymmetry and expand meaningful data access for both patients and independent analytic services. Interoperability must move beyond limited, summary-level report exchange to full, usable datasets. AI cannot improve outcomes in a system where transparency and data portability are limited.

2. Increase Patient Choice in Diagnostic Interpretation

The current system offers limited patient choice in diagnostic interpretation. While developments such as direct patient lab ordering have improved access, diagnostic imaging services remains largely closed. Given the absence of or decrease in independent diagnostic imaging services, HHS should encourage models that decouple radiology data collection from interpretation. Patients should be able to choose independent clinically validated AI tools to interpret their images — including tools developed by technology companies operating outside vertically integrated health systems.

Competition at the interpretive layer will provide more useful information to patients, spur innovation, and improve quality.

3. Recognize Radiology as a Precision Medicine Data Asset

Radiology imaging acquisition is fundamentally a complex data collection tool. Interpretation should move beyond narrative summaries that answer a narrow clinical question and instead produce structured anatomical and pathological inventories.

Imaging providers should compete not on volume, but on the quality and completeness of their diagnostic outputs. Structured, standardized inventories of anatomy and pathology rather than a brief or incomplete answer to a narrow clinical question noted on a report can serve as reusable groundwork for precision medicine across time, providers, and analytic tools.

Standards should promote:

- Structured data capture
- Longitudinal comparability
- Reusability for AI training
- Precision treatment optimization

Radiology data should be recognized as a reusable asset owned by the patient. Currently, systems acquire, interpret, manage, and control or restrict access to imaging data— and then deliver or not limit

treatment based on that controlled or limited dataset. This can function as gatekeeping and limit patient choice.

A competitive payment structure should allow independent AI companies to use acquired imaging data to interpret radiology imaging or other tests for better outcomes. This will spur innovation in diagnostic timeliness and quality, structured reporting standards, and precision-based therapies.

4. Tie AI Incentives to Measurable Outcomes

AI payment incentives should reward:

- Reduced time-to-diagnosis
- Fewer missed diagnoses
- Improved morbidity and mortality outcomes

AI policy should stimulate competition that measurably improves diagnostic accuracy while improving efficiency to reduce costs. For example, cardiovascular disease remains one of the highest-cost areas in Medicare and is largely diagnosed and managed through imaging. More complete, detailed, and accurate image interpretation can lead to earlier intervention, lower downstream costs, and improved outcomes.

What are the most promising AI evaluation methods? Should HHS support these processes?

1. The most promising approach is comparative, outcome-based evaluation in real-world settings — especially for radiology imaging services. Evaluation should include:
 - Institutional vs. independent AI interpretation comparisons
 - Divergence analysis in abnormality detection
 - Time-to-diagnosis tracking
 - Missed pathology rates
 - Downstream outcome measurement
2. Evaluation must address data integrity—an underrated and underdeveloped area.

AI underperformance is often driven by poor input data rather than algorithmic weakness. An AI radiology system that reviews imaging, labs, pathology, and notes must also incorporate patient-generated data. Provider documentation and diagnostic test results frequently:

- Omits or misstates relevant history
- Contains incomplete or incorrect clinical orders
- Maintains outdated or inaccurate diagnoses
- Fails to correlate imaging findings with patient-reported symptoms

If AI systems are trained or evaluated on distorted documentation, performance metrics will be misleading. HHS should prioritize research on:

- Patient-facing radiology AI tools

- Interfaces embedded in medical records with structured symptom tracking
- Integration of device-generated data (e.g., blood pressure)
- Comparative accuracy studies with and without patient-reported context

Including structured patient context can meaningfully improve diagnostic accuracy and clinical decision-making.

Where would enhanced interoperability accelerate development?

1. Interoperability is most critical in imaging diagnostics. It should ensure:
 - Full DICOM portability without compression
 - Structured linkage between imaging, labs, pathology, and longitudinal records
 - Integration of patient-reported symptoms
 - Provenance and audit tracking across systems/users
 - Historical imaging comparison capabilities, rather than just the reports
2. Radiology archives contain significant underutilized diagnostic capacity. Radiology imaging studies hold far more signal than narrative reports capture. Market opportunity expands when:
 - Independent analytic services can access standardized datasets
 - Hospitals compete on the structure and completeness of diagnostic inventories
 - Patients can submit imaging for competitive interpretation
 - Researchers can benchmark models on complete, traceable data
3. Patients should be able to choose which AI tools to embed in their records and present AI-generated summaries at visits. This supports collaborative, informed decision-making rather than unilateral interpretive control.

What challenges do patients want addressed? What concerns exist?

1. Patients primarily seek:
 - Timely and accurate diagnosis
 - Earlier detection of serious conditions
 - Accurate and complete medical records
 - Detailed radiology reports that correlate findings with symptoms and longitudinal history
 - Transparency about AI tools used and access to their outputs

Clinical AI must align with patient needs — not solely clinician workflows. Excluding patients as a primary stakeholder risks embedding inefficiencies and unsafe assumptions into new systems.

2. A central but longstanding challenge is documentation distortion. For example: A patient with an indeterminate liver lesion who recently experienced a serious cardiovascular event that was not properly documented. Without the patient context, an AI tool may mischaracterize the lesion. Similarly, if the imaging order reflects an inaccurate problem framing, interpretation may be misdirected. Unless AI incorporates structured patient-reported history and symptom tracking,

outputs will be incomplete or inaccurate. The strength of AI accuracy and efficiency is directly related to the quality and degree of information it consumes.

3. Patients are concerned about:

- AI reinforcing inaccurate notes and outdated diagnostic history
- AI trained on incomplete or uncorrelated imaging reports
- Continued exclusion from interpretive processes
- Systems optimized for administrative efficiency rather than clinical outcomes
- Lack of transparency about which AI tools are used

The quality of the medical record directly determines the reliability of AI tools that affect patient outcomes.

What AI research should HHS prioritize?

1. HHS should prioritize direct-care, patient-facing radiology imaging interpretation tools that:

- Analyze full imaging datasets rather than narrative summaries
- Produce structured anatomical and pathology inventories
- Correlate findings with patient history
- Fully and clearly disclose abnormalities
- Support cardiovascular imaging interpretation

Radiologist shortages and increasing imaging complexity make innovation urgent.

2. HHS should also prioritize research on patient-integrated contextual AI systems that:

- Review imaging, labs, pathology, notes, and patient-reported data simultaneously
- Identify inconsistencies between documentation and history
- Flag divergence
- Generate structured differential summaries

3. Studies should measure diagnostic accuracy and outcomes with and without integrated patient-reported context.

Radiology data contains substantial unused diagnostic capacity. AI policy should unlock that capacity through transparency, structured data standards, competition for quality, and alignment with patient needs — not merely digitize existing workflows and limited outputs.

Thank you for this opportunity to comment.